

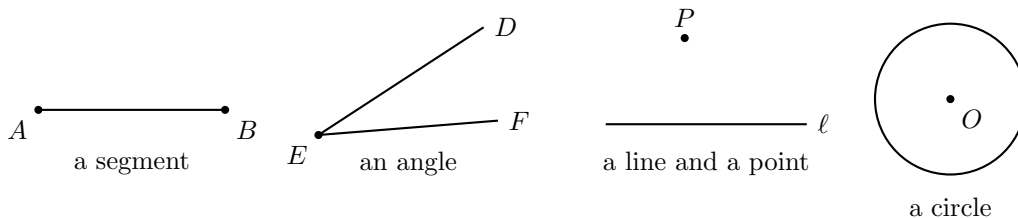
Rigid Motions and Constructions — Planning and Justifying Compass-and-Straightedge Work

High School Geometry

Writing a construction as an ordered plan, justifying it with the congruent triangles the arcs create, and checking the finished claim on coordinates

Directions:

- A **plan** is a numbered list of moves. Each step must say where the compass point sits, whether the opening changes, and what gets drawn. “Draw an arc” is not a step until it says *centred where* and *with what radius*.
- Legal tools only: an unmarked straightedge and a compass. No ruler measurements, no protractor.
- A **justification** names the congruent triangles or the equal radii that force the result, and the criterion (SSS, SAS, ASA) or definition being used. “It looks right” justifies nothing.
- The **coordinate check** problems do not ask you to construct anything. They hand you the output of a construction and ask you to test the property that construction is supposed to guarantee. Say which property you are testing before you compute, and round lengths to the nearest hundredth.
- The figures below are for reference only. They carry no measurements: use the letters, and take any numbers from the problem you are on.



Reference figures. No values shown: use the letters here and the numbers given in each problem.

1. Write the plan for constructing the **perpendicular bisector** of segment AB with compass and straightedge.

Number your steps. For every arc, say where the compass point sits and whether the opening changed from the step before. Then state the one condition the opening must satisfy for the construction to work at all, and say what goes wrong if it does not.

2. Write the plan for constructing the **bisector of angle DEF** .

Number your steps. Mark clearly which two arcs must share the same compass opening, and say why the construction fails if they do not.

3. A perpendicular bisector was constructed on segment AB , where $A(-2, 1)$ and $B(6, 7)$. The construction produced the point $P(-1, 8)$.

Name the property a perpendicular bisector guarantees about P , then test it: compute PA and PB .

Answer: _____

4. Justify the perpendicular-bisector construction. In the finished figure the two arc intersections are called M and N , and the endpoints of the segment are A and B .

(a) Name two congruent triangles in that figure and the criterion that makes them congruent, saying exactly which lengths are equal *by construction*.

(b) Explain how that congruence forces the line MN to be both perpendicular to AB and to pass through its midpoint.

5. A regular hexagon is inscribed in a circle of radius 10 centred at the origin, by walking the compass — still open to the radius — around the circle. Two consecutive vertices land at $(10, 0)$ and $(5, 8.66)$.

State the claim that construction depends on, then test it by computing the distance between those two vertices.

Answer: _____

6. Write the plan for constructing the line through point P that is **parallel to line ℓ** , using the copy-an-angle method.

Number your steps. Say which step creates the pair of corresponding angles, and say why a protractor may not be used to make the copy.

7. A segment AB with $A(1, 2)$ and $B(7, 10)$ is copied onto a ray starting at $C(3, -4)$ by opening the compass to AB and swinging that opening from C . The copy lands with endpoint $D(11, 2)$.

Name the property the copy construction guarantees, then test it: compute AB and CD .

Answer: _____

8. Justify the equilateral-triangle construction. Given segment AB , you open the compass to AB , draw an arc centred at A , draw a second arc centred at B with the *same* opening, and call their intersection C .

Prove that triangle ABC is equilateral. Your argument must use the definition of a circle, not the appearance of the drawing.

9. The perpendicular bisectors of two sides of the triangle with vertices $A(0,0)$, $B(8,0)$ and $C(0,6)$ were constructed, and they meet at $O(4,3)$.

State what the intersection of two perpendicular bisectors is guaranteed to be, then test it: compute OA , OB and OC .

Answer: _____

10. Challenge. Given segment AB , plan a construction of the point Q on AB with AQ equal to one quarter of AB .

- (a) Write the full plan, numbered.
- (b) Justify it: for each bisection you use, name the congruence or equidistance argument that guarantees the point you get really is a midpoint.
- (c) A classmate says the same repeated-bisection idea gives a way to trisect any segment. Explain why it does not.